

# **Oulu Bus Network GTFS Analysis — Full Case Study Report**

**Version: 1.0**

**Date: June 2026**

***Following the structure of “Case Study 3: Follow your own case study path” (Google Data Analytics Professional Certificate)***

## **ASK**

### **Business Task**

This is a self-initiated portfolio project created as part of the Google Data Analytics Professional Certificate. The goal is to apply the analytical skills learned in the program to a real-world dataset relevant to the city where I study (Oulu, Finland).

The project simulates a realistic consulting scenario in which a data analyst evaluates the performance of the Oulu public bus network using GTFS data. The objective is to identify route efficiency, stop usage patterns, and opportunities for improving public transport service planning.

### **Key Factors**

Key factors influencing the analysis include:

- route structure
- trip frequency and scheduling
- stop usage patterns
- geographical distribution of stops and routes
- service availability on different days

These factors affect passenger experience, network efficiency, operational costs, and service reliability.

### **Data Needed**

The analysis requires GTFS (General Transit Feed Specification) data, including:

- stops
- routes
- trips
- stop-level arrival/departure times

- service calendars
- route shapes
- fare information

### **Data Source**

The dataset was obtained from the official Waltti Open Data portal (<https://opendata.waltti.fi/docs>), which provides GTFS feeds for public transport systems across Finland, including Oulu. Files used include:

- stops.txt
- routes.txt
- trips.txt
- stop\_times.txt
- calendar.txt
- calendar\_dates.txt
- shapes.txt
- fare\_attributes.txt
- fare\_rules.txt

### **Audience**

- transportation planners and mobility analysts
- BI and data analytics professionals
- potential employers
- academic supervisors

### **Business Task Statement**

Analyze the performance of the Oulu public bus network using GTFS data to identify the most heavily used routes and stops, evaluate schedule efficiency, detect bottlenecks, and provide data-driven recommendations for improving service planning.

## **PREPARE**

### **Data Location**

The dataset is stored locally in a project folder Oulu\_Bus\_Network\_2026, with original GTFS files placed in data\_raw/. This structure is mirrored in the GitHub repository.

## Data Organization

The dataset follows the GTFS standard. Core files include:

- agency.txt
- routes.txt
- stops.txt
- trips.txt
- stop\_times.txt
- calendar.txt
- calendar\_dates.txt
- shapes.txt

Additional files (not required for core analysis):

- fare\_attributes.txt
- fare\_rules.txt
- transfers.txt
- vehicle\_groups.txt
- contracts.txt
- emissions.txt

## ROCCC Evaluation

The dataset meets ROCCC criteria:

- **Reliable** — published by Oulu public transport
- **Original** — sourced directly from Digitraffic
- **Comprehensive** — includes all GTFS components
- **Current** — regularly updated
- **Cited** — publicly documented

No personal or sensitive data is included.

## Licensing & Privacy

- GTFS feed is open and publicly available
- No PII
- Only aggregated insights used

### **Data Integrity Verification**

I verified:

- presence of all required GTFS files
- valid headers
- correct delimiters
- valid relationships (route\_id, trip\_id, stop\_id)
- correct date formats
- valid geographic coordinates

### **Data Limitations**

- no passenger counts
- no real-time delays
- some files not needed for analysis
- large files require SQL/Python for processing

### **Deliverable Summary**

The GTFS dataset is structurally sound, complete, and ready for analysis.

## **PROCESS**

### **Tools Used**

- **Excel** — initial inspection
- **SQL (BigQuery)** — analysis of large files
- **Tableau** — visualizations

### **Data Integrity Steps**

- verified headers
- checked missing values
- validated IDs

- confirmed expected empty columns
- checked date formats

### **Cleaning Steps**

No major cleaning required; GTFS is already standardized. Only inspection and validation were performed.

### **Deliverable Summary**

The dataset was inspected, validated, and confirmed ready for analysis.

## **ANALYZE**

### **Data Organization**

I organized SQL scripts into folders:

- exploration
- joins
- summaries
- trends
- table creation

### **Formatting Check**

GTFS tables were consistent and required no reformatting.

### **Unexpected Findings**

- arrival\_time values above 24 hours (GTFS standard for post-midnight trips)
- uneven distribution of trips across routes and hours

### **Trends Identified**

- some routes have significantly more trips
- certain stops act as major hubs
- longest routes differ from routes with most trips
- peak activity occurs during morning and afternoon hours
- **Monday has the highest service level; all other days are identical**
- late-night activity exists (24–28 hours)

## **How Insights Answer the Business Task**

Insights reveal:

- busiest routes
- most important stops
- peak demand periods
- weekday patterns

These support decisions about scheduling, resource allocation, and service optimization.

### **Deliverable Summary**

The analysis revealed clear patterns in route usage, stop popularity, peak hours, and weekday service levels.

## **SHARE**

### **Findings**

- busiest routes and stops identified
- peak travel hours visualized
- Monday stands out as the only day with increased service
- late-night activity highlighted

### **Visualizations Created**

- Peak Hours
- Trips per Route
- Longest Routes
- Top Stops
- Weekday Services

### **Deliverable Summary**

Visualizations clearly communicate key insights and support stakeholder decision-making.

## **ACT**

## **Final Conclusion**

The Oulu bus network shows clear usage patterns: Monday has the highest service level, while all other days follow an identical pattern. Peak hours occur between 8:00 and 17:00, and a small number of routes and stops carry most of the demand.

## **How Insights Can Be Applied**

- optimize scheduling
- increase frequency on high-demand routes
- evaluate low-usage routes
- plan for peak-hour capacity
- improve service reliability

## **Next Steps**

- analyze seasonal trends
- incorporate passenger counts
- include real-time delay data
- evaluate vehicle capacity

## **Top High-Level Insights**

- Monday has the highest service level; all other days are identical
- peak hours 8:00–17:00
- few routes carry most trips
- major hub stops identified
- late-night service exists (24–28 hours)